TC-ViTok: A Failure-Mode Taxonomy for Pre-LLM Visual Token Compression, with a Parameter-Free Recipe

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Abstract

We report a trained, pre-LLM, query-conditioned visual-token selector (TC-ViTok) at LLaVA-1.5-7B scale with a matched same-LoRA, same-SFT, full-576-token reference (LoRA-576). At $4\times$ compression (measured as the $k{=}144$ selected-token ratio), TC-ViTok incurs a measured cost of -4.1 VQAv2, -3.4 GQA, and -7.4 POPE points — a cost we report rather than hide.

Two empirical observations follow from this comparison. First, the converged selector is empirically close to query-agnostic: zeroing the CLIP query embedding at inference produces equal or marginally better accuracy, indicating that the architecture's query-routing channel is not used strongly at this scale. Second, the converged selector exhibits a structured per-position prior with a localized region whose inference-time masking improves exact-match VQA-score by $+2.2$ pp on Final, ranks at the 90th percentile of 10 size-matched random-rectangle controls ($p{=}0.0107$, one-tailed sign test). The effect is positive on two related checkpoints but reverses sign on an independently retrained seed, so we treat it as a low-utility-allocation diagnostic rather than a checkpoint-invariant mechanism.

To explain why compression does not close the LoRA-576 gap, we document three training-dynamics pathologies we encountered while training a query-conditioned visual-token selector co-optimized with LLM adapters before the LLM. We present them as concrete, testable hypotheses about pre-LLM trainable compression more broadly; their universality is an open question we explicitly invite future work to resolve. *Co-adaptation oscillation* produces limit cycles whose period grew from approximately 5K to 15K steps as the selector learning rate dropped $25\times$ across the two finite-period measurements available (the frozen-pooler control has no cycle). We report this as a qualitative trend, not a scaling law; two data points are insufficient to fit a functional form. *BudgetPredictor collapse* sends per-sample token budgets to $k_{\max}$ whenever $\lambda_b \leq 0.05$, with a narrow stable window of $[0.08, 0.15]$. *Warm-start dependence* prevents from-scratch convergence at LLaVA-1.5 scale across two selector variants and three LoRA capacities. We sidestep each pathology with a parameter-free token merge, a calibrated λ_b, and a staged curriculum on a converged base.

1. Introduction

Pre-LLM visual token compression — reducing the visual sequence before it enters the LLM, saving all prefill FLOPs — does not naively converge when the selector is co-trained with LLM adapters. This paper is about why, and what to do about it.

Modern VLMs such as LLaVA-1.5 (Liu et al., 2023) attach a large language model (LLM) to a frozen CLIP ViT-L/14 vision encoder. At inference, the encoder maps a 336×336 image to 576 patch tokens (24×24 grid), which are prepended to the tokenized question and fed into the LLM. Since attention cost is quadratic in sequence length, these 576 visual tokens dominate prefill FLOPs — and the budget is determined by image resolution, not by what the question requires.

Prior work compresses tokens either query-agnostically (ToMe (Bolya et al., 2023), EViT (Liang et al., 2022), FastV (Chen et al., 2024), PruMerge (Shang et al., 2024)) or inside the LLM's attention layers, after all tokens have been injected (SparseVLM (Zhang et al., 2025), Focus-LLaVA (Zhu et al., 2024)). Pre-LLM compression with a learned query-conditioned selector is the architecturally natural design: it saves LLM FLOPs and is modular. It also does not converge in the obvious training setup.

We attempted this design and found that the obvious joint training procedure fails in three coupled ways: co-adaptation oscillation, budget-head collapse, and warm-start dependence. We characterize each failure quantitatively, derive an architectural recipe that sidesteps two of them by construction and characterizes the third empirically, and report an honest compression cost against the matched same-LoRA baseline (LoRA-576). TC-ViTok's role in the paper

is to make these failure modes observable under a realistic LLaVA-scale training setup.

We emphasize that TC-ViTok loses to a matched LoRA-576 no-compression reference by 4.1 / 3.4 / 7.4 points on VQAv2 / GQA / POPE. The paper's contribution is the characterization of why pre-LLM trainable compression fails to close this gap, not the method that fails to.

Practically, the paper is most useful as a map. If future work attempts trainable pre-LLM token compression at LLaVA-1.5-7B scale, our results suggest it should budget for co-adaptation oscillation, budget-head collapse, and warm-start dependence, and should test whether the converged selector is actually using the query signal rather than assuming architectural query-conditioning is realized by training.

We make three contributions.

(1) A failure-mode taxonomy for pre-LLM trainable token compression. Three coupled pathologies arise in our setting when a learnable selector is jointly trained with LLM adapters: (a) *GAN-style co-adaptation oscillation* whose period increases as selector learning rate decreases in the measured trainable-pooler family; (b) *BudgetPredictor collapse* to $k{=}k_{\max}$ whenever $\lambda_b \leq 0.05$; and (c) *warm-start dependence* — from-scratch attempts plateau 2–10$\times$ above the warm-started ceiling regardless of TPN variant (Section 6.2, Table 9). All evidence for this taxonomy comes from one codebase and one (encoder, LLM, selector) triple; we present the taxonomy as a set of portable hypotheses, not as a universality theorem.

(2) An architectural recipe that sidesteps two failures and characterizes the third. (i) A tau-independent additive STE correction (Eq. 7) that keeps TPN-score gradients non-zero at any annealed temperature; (ii) *Informative Token Merge*, parameter-free by design, which eliminates the co-adaptation loop by construction — removing the trainable bridge removes the closed gradient circuit; and (iii) a *TPN warmup → joint warm-freeze → adaptive budget* curriculum whose stage ordering is the only ordering of TPN-warmup, joint-freeze, and adaptive-budget stages we tested that achieves convergence; exhaustive testing of alternative orderings is left to future work. The warm-start dependence (FM3) is characterized empirically but not defeated: from-scratch training does not converge at this scale regardless of architecture or curriculum order.

(3a) Empirical query-agnostic convergence at LLaVA-7B scale. The trained query-conditioned TPN, evaluated on three benchmarks at $k{=}144$, produces equal or marginally better accuracy when the CLIP query embedding is zeroed at inference (E3, §6.7 L7; $+0.40$ / $+0.33$ / $+0.28$ pp on VQAv2 / GQA / POPE). On Adaptive v2, the same 5K-sample protocol gives $+0.56$ / -0.16 / $+0.41$ pp, so zeroing the query again produces no material degradation. This is an unexpected property of joint optimization at this scale, and to our knowledge has not been reported in prior pre-LLM compression work.

(3b) Structured per-position prior with a localized low-utility region. The converged selector exhibits strong per-position selection-frequency heterogeneity (Fig. 11) and a localized region whose inference-time masking improves accuracy at the 90th percentile of random controls (M1–M3, §6.7 L9). The direction is positive on Final, Adaptive v2, and v3, but negative on an independently retrained seed-7 checkpoint; this makes L9 an observational low-utility-allocation finding rather than a claim of checkpoint-invariant dead-zone causality.

2. Related Work

Query-agnostic visual token compression. ToMe (Bolya et al., 2023) merges visually similar tokens via bipartite matching inside ViT layers, achieving 2$\times$ throughput at modest accuracy loss on classification. DynamicViT (Rao et al., 2021) learns binary dropping masks at intermediate layers via Gumbel-Softmax, trained with distillation. EViT (Liang et al., 2022) uses CLS-token attention to rank tokens and fuses low-importance ones. Post-encoder methods include LLaVA-PruMerge (Shang et al., 2024), which prunes by CLS-attention rank then merges pruned tokens with nearest retained neighbors via key-similarity weighting. FastV (Chen et al., 2024) observes that LLM text-to-visual attention collapses after layer 2 and prunes low-attention tokens at that boundary. All these methods share a fundamental limitation: the compression decision is made without knowledge of the query, systematically harming spatially-grounded tasks (localization, OCR) more than global tasks (Chen et al., 2024; Endo et al., 2025).

Query-aware compression. BLIP-2 (Li et al., 2023a)'s Q-Former extracts visual tokens via cross-attention with learned query embeddings, but the queries are fixed parameters shared across all inputs rather than derived from the user's question. SparseVLM (Zhang et al., 2025) incorporates text guidance through the LLM's own text-to-visual attention with adaptive per-layer recycling, but selection happens inside the LLM after all tokens have already been injected. QG-VTC (Li et al., 2025) achieves near-uncompressed performance with 1/8 of tokens by embedding the question into the ViT's feature space and progressively compressing within the encoder — but this couples compression to the encoder's internal architecture. FocusLLaVA (Zhu et al., 2024) applies coarse-then-fine instruction-guided selection, but the text-guided stage remains inside the LLM. TC-ViTok closes this gap by implementing query-conditioning as a modular pre-LLM module that operates on frozen, pre-computed features from both encoders.

Table 1. Positioning of TC-ViTok in the visual token compression landscape. Pre-LLM = decision made before LLM forward pass (saves all LLM FLOPs); inside LLM = decision made between LLM layers (saves only late-layer FLOPs).

Method	Where	Query	Recover	Adapt k
ToMe (Bolya et al., 2023)	in-encoder	✗	merge	✗
DynamicViT (Rao et al., 2021)	in-encoder	✗	✗	✗
EViT (Liang et al., 2022)	in-encoder	✗	fuse	✗
PruMerge (Shang et al., 2024)	post-encoder	✗	merge	✗
FastV (Chen et al., 2024)	in-LLM (L=2)	implicit	✗	✗
SparseVLM (Zhang et al., 2025)	in-LLM	via attn	recycle	per-layer
QG-VTC (Li et al., 2025)	in-encoder	explicit	merge	✗
FocusLLaVA (Zhu et al., 2024)	in-LLM	explicit	coarse-fine	✗
TC-ViTok (ours)	**pre-LLM**	**explicit**	**merge**	**per-sample**

Differentiable discrete optimization. The Gumbel-Softmax trick (Jang et al., 2017; Maddison et al., 2017) enables gradient estimation through categorical variables via a continuous relaxation parameterized by temperature. The straight-through estimator (STE) (Bengio et al., 2013) uses the hard binary mask in the forward pass while routing gradients through the soft relaxation, providing low-variance gradient estimates for subset selection. AdaViT (Meng et al., 2022) studies learned per-sample computation budgets in ViTs. TC-ViTok extends these ideas with (i) Gumbel-Sigmoid for independent binary decisions across 576 positions, and (ii) an additive tau-independent gradient correction that resolves a gradient vanishing failure mode at low temperature absent from prior work.

Table 1 positions TC-ViTok in this landscape along four operationally-relevant axes: *when* compression occurs, *whether* the question conditions selection, *whether* discarded tokens are recovered, and *whether* the budget can vary per sample.

To our knowledge, TC-ViTok is the first method that combines (a) explicit question-conditioned selection, (b) pre-LLM placement on frozen features, and (c) per-sample adaptive budget into a single modular system.

3. Methodology

3.1. Problem Formulation

Given an image $\mathbf{I}$ and a natural-language question $\mathbf{Q}$, a VLM f_θ generates an answer $\mathbf{A}$. Standard LLaVA-1.5 encodes $\mathbf{I}$ into $N = 576$ patch tokens $\mathbf{v}_f \in \mathbb{R}^{N\times D_v}$ (where $D_v = 1024$) and processes all of them. We instead learn a policy:

$$\pi_\phi(\mathbf{I}, \mathbf{Q}) \;\to\; \mathcal{S} \;\subseteq\; \{1, \ldots, N\}, \quad |\mathcal{S}| = k$$

where $k \ll N$ and ϕ is trained end-to-end from the VQA loss. The LLM receives only the k selected tokens, directly reducing prefill FLOPs.

3.2. The Pre-LLM Co-Adaptation Problem

Before describing the architecture, we state the structural problem that motivates the design. Modular pre-LLM compression places a learnable selector π_ϕ between the visual encoder and an LLM whose adapters θ_{LoRA} are also trained. Denote the LLM input distribution $p_{\text{LLM}}(\mathbf{v}_{sel}) = \pi_\phi(\mathbf{v}_f)$ and the LLM output distribution $p_\theta(y|\mathbf{v}_{sel}) = f_{\theta_0+\theta_{\text{LoRA}}}(\mathbf{v}_{sel})$. The shared VQA loss $\mathcal{L}(\phi, \theta_{\text{LoRA}}) = \mathbb{E}_{(I,Q,y)}[-\log p_\theta(y \mid \pi_\phi(I, Q))]$ then induces a closed gradient circuit:

$$\underbrace{\mathcal{L}}_{\text{task loss}} \xrightarrow{\nabla_\theta} \underbrace{\Delta\theta_{\text{LoRA}}}_{\text{LoRA update}} \to \underbrace{\text{shifted } p_\theta(\cdot|\mathbf{v}_{sel})}_{\text{LLM "moves"}} \to \underbrace{\nabla_\phi\mathcal{L}}_{\text{new selector grad}} \to \underbrace{\Delta\phi}_{\text{selector upd}}$$

Both parameter groups respond to a moving target. This is structurally identical to a non-saturating two-player game; the equilibrium analysis of GAN-style optimization (Mescheder et al., 2018; Jolicoeur-Martineau, 2018) predicts limit cycles whose period is set by the slower learner. Empirically (Section 6.1, Fig. 13), this manifests as oscillatory val-VQA trajectories with period $T_{\text{osc}} \approx c/\eta_{\text{selector}}$ for an architecture-dependent constant c ($c \approx 2.5$ for our trainable-pooler variants); the relation holds across eight architecturally distinct trainable-pooler variants and was observed down to $\eta_{\text{selector}}{=}2{\times}10^{-5}$, where the period extended to ≈15K steps but the cycle persisted.

This motivates three design choices that pervade the rest of the methodology, each addressing one edge of the gradient loop. *First*, the recovery of unselected tokens is *parameter-free* (Section 3.6): with no learnable bridge between the selector and the LLM, one edge of the cycle is structurally absent and the loop cannot close. *Second*, the optimizer schedule warm-freezes the LoRA adapters for the first 3000 steps so the selector reaches a stable operating regime before any LLM adaptation begins (breaking the loop *in time*). *Third*, the adaptive-budget stage trains only on a converged base checkpoint (Section 6.2); from-scratch joint training of a high-capacity selector + LoRA pair plateaus 2–10× above the warm-started ceiling regardless of TPN variant, learning rate, or diversity-loss weight (Section 6.2, Table 9). All three decisions are derived from, not prior to, the failure analysis in Sections 6.1–6.2.

3.3. Frozen Encoders

We use three frozen pre-trained encoders. **DINOv2-S** (Oquab et al., 2023) encodes a 98×98 crop of $\mathbf{I}$ (7 patches per side) to produce a coarse spatial map $\mathbf{v}_c \in \mathbb{R}^{49\times384}$. We use 98px input (not the standard 518px) to give the TPN a cheap, 7×7 spatial grid that is efficient to cross-attend. DINOv2's self-supervised training preserves patch-level spatial locality better than CLIP's contrastive objective, making it a more precise spatial guide. **CLIP ViT-B/32** (Radford et al., 2021) encodes $\mathbf{Q}$ to produce a query embedding

$\mathbf{q} \in \mathbb{R}^{512}$ (the EOS token representation). **CLIP ViT-L/14** (Radford et al., 2021) encodes a 336×336 crop to produce full-resolution features $\mathbf{v}_f \in \mathbb{R}^{576\times1024}$, which are already aligned to LLaVA-1.5's LLM since LLaVA uses the same encoder. All three encoders are always frozen — no gradients flow into them.

3.4. Tokenization Policy Network (TPN)

The TPN (≈887K parameters) maps $(\mathbf{v}_c, \mathbf{q})$ to importance scores $\mathbf{s}_N \in \mathbb{R}^N$ and selection indices $\text{idx} \in \mathbb{Z}^k$.

Cross-attention (4 heads). The question embedding is treated as a single query:

$$Q = \mathbf{q}\,W_Q \in \mathbb{R}^{1\times384}, \quad K = \mathbf{v}_c\,W_K \in \mathbb{R}^{M\times384}, \quad V = \mathbf{v}_c\,W_V \in \mathbb{R}^{M\times384}$$

where $M = 49$. Multi-head attention yields an attended context vector $\text{ctx} \in \mathbb{R}^{384}$ and head-averaged spatial weights $\boldsymbol{\alpha} \in \mathbb{R}^M$. The context encodes *what* visual content is relevant to the question; $\boldsymbol{\alpha}$ encodes *where*.

Score MLP. $[\text{ctx};\, \boldsymbol{\alpha}] \in \mathbb{R}^{433}$ is passed through a LayerNorm and two-layer MLP to produce 49 coarse scores, then bilinearly upsampled $7 \times 7 \to 24 \times 24$ to yield $\mathbf{s}_N \in \mathbb{R}^{576}$.

Instance normalization. $\mathbf{s}_N$ is normalized to zero mean and unit variance per sample: $\mathbf{s}_N \leftarrow (\mathbf{s}_N - \mu)/\sigma$. Without this, Adam pushes score magnitudes to ±100 within $\sim$1000 steps. At that scale, Gumbel noise (magnitude $\approx \tau \approx 1.0$) is negligible compared to scores, and the same 576 positions are always selected regardless of question — entropy collapse. Alternative normalizations (tanh, batch norm) also fail: tanh saturates at ±1 ($|\tanh'(\pm100)| \approx 10^{-87}$), and batch norm carries a stateful running mean that disrupts inference. Instance norm permanently anchors scores to $\mu = 0, \sigma = 1$ with no learnable parameters the optimizer can circumvent.

3.5. Gumbel-Sigmoid Straight-Through Estimator

Token selection is a discrete top-k operation with zero gradients almost everywhere. We adapt the Gumbel-Sigmoid STE (Jang et al., 2017; Bengio et al., 2013) for independent binary subset selection over 576 positions.

Forward pass. Gumbel noise $\mathbf{g} \sim \text{Gumbel}(0, 1)$ provides per-sample diversity; the temperature τ controls the soft-to-hard transition:

$$\mathbf{n} = \mathbf{s}_N + \mathbf{g}, \tag{1}$$

$$\mathbf{m}_{\text{soft}} = \sigma(\mathbf{n}/\tau) \in (0, 1)^N, \tag{2}$$

$$\text{idx} = \text{topk}(\mathbf{n},\, k), \tag{3}$$

$$\mathbf{m}_{\text{hard}} = \mathbf{0};\ \mathbf{m}_{\text{hard}}[\text{idx}] = 1, \tag{4}$$

$$\mathbf{m}_{\text{STE}} = \mathbf{m}_{\text{hard}} - \mathbf{m}_{\text{soft}}.\text{detach}() + \mathbf{m}_{\text{soft}}. \tag{5}$$

In the forward pass, $\mathbf{m}_{\text{STE}} = \mathbf{m}_{\text{hard}}$ (the $\mathbf{m}_{\text{soft}}$ terms cancel). In the backward pass, the gradient flows through $\mathbf{m}_{\text{soft}}$.

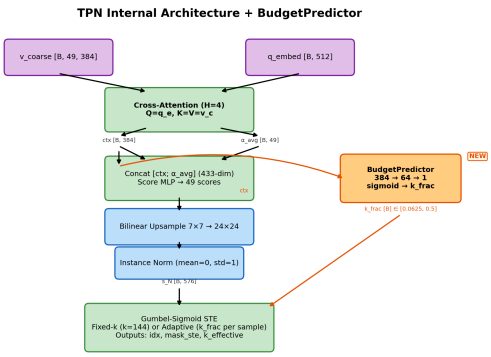


Figure 1. **TPN internal architecture and BudgetPredictor wiring.** Cross-attention with $H{=}4$ heads ($Q{=}\mathbf{q}$, $K{=}V{=}\mathbf{v}_c$) produces a context vector $\text{ctx}{\in}\mathbb{R}^{384}$ and head-averaged spatial weights $\boldsymbol{\alpha}{\in}\mathbb{R}^{49}$. The $[\text{ctx};\boldsymbol{\alpha}]$ concatenation passes through a Score MLP producing 49 coarse scores, which are bilinearly upsampled $7{\times}7 \to 24{\times}24$ and instance-normalized to yield $\mathbf{s}_N{\in}\mathbb{R}^{576}$. Phase 1 Adaptive shares ctx with the BudgetPredictor (orange) which emits the per-sample token fraction $k_{\text{frac}}{\in}[0.0625, 0.5]$ consumed by the Adaptive Gumbel-STE.

Noise in Eq. (1) (not raw scores) is used for top-k to ensure per-sample selection diversity across the batch; using raw scores selects identical positions for all samples, collapsing entropy. Temperature is annealed linearly $\tau : 1.0 \to 0.3$ over training.

Tau-independent additive gradient correction. At low τ, $\mathbf{m}_{\text{soft}}$ saturates toward $\{0, 1\}$, making $\partial\mathbf{m}_{\text{soft}}/\partial\mathbf{s}_N \approx 0$ for all positions with $|\mathbf{s}_N[i]| > 1$. This kills the gradient path to TPN weights, halting learning after τ has annealed. We resolve this by multiplying gathered visual tokens by an additive correction:

$$\mathbf{v}_{sel} = \text{gather}(\mathbf{v}_f, \text{idx}), \quad \mathbf{c} = \text{gather}(\mathbf{s}_N, \text{idx}), \tag{6}$$

$$\mathbf{v}_{sel} \leftarrow \mathbf{v}_{sel} \odot \big(1 + \mathbf{c} - \mathbf{c}.\text{detach}()\big). \tag{7}$$

In the forward pass, $\mathbf{c} - \mathbf{c}.\text{detach}() = \mathbf{0}$, so $\mathbf{v}_{sel}$ is unchanged. In the backward pass, for selected position j:

$$\frac{\partial\mathcal{L}_{\text{VQA}}}{\partial\mathbf{s}_N[j]} = \sum_{d=1}^{D_v} \frac{\partial\mathcal{L}}{\partial\mathbf{v}_{sel}[j,d]} \cdot \mathbf{v}_{sel}[j,d],$$

a dot product between the VQA gradient and visual features at position j. This gradient is non-zero at any τ and provides a dense, token-level supervision signal for TPN throughout training.

3.6. Informative Token Merge

Hard dropping of the 432 unselected tokens discards all the contextual information they carry. PruMerge (Shang

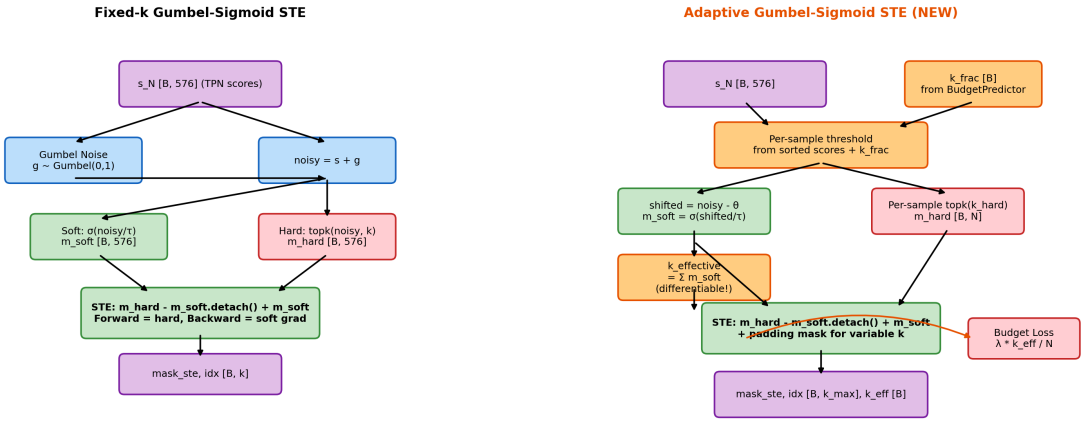


Figure 2. **Fixed-k vs. Adaptive Gumbel-Sigmoid STE** (Phase 1 Final vs. Phase 1 Adaptive). *Left:* the standard fixed-budget pipeline: scores + Gumbel noise feed both a soft sigmoid (for backward pass) and a hard top-k=144 selection (for forward pass), composed via the STE identity $\mathbf{m}_{\text{STE}}=\mathbf{m}_{\text{hard}}-\mathbf{m}_{\text{soft}}.\text{detach}()+\mathbf{m}_{\text{soft}}$. *Right:* the adaptive variant inserts a per-sample threshold θ derived from the BudgetPredictor's k_{frac} output. The shifted soft mask $\sigma((\mathbf{n}-\theta)/\tau)$ produces a differentiable effective count $k_{\text{eff}}=\sum_i m_{\text{soft},i}$ that drives the budget loss; per-sample top-k_{hard} provides the hard mask with right-padding to $k_{\max}$ for batched processing.

et al., 2024) shows that merging such tokens recovers ∼2% VQAv2 over pure pruning at the same budget. We implement a query-conditioned variant: the TPN score of an unselected token determines how strongly it contributes to the merge, so tokens that were "almost selected" (near-threshold, task-relevant) contribute more than distant background tokens.

Let $\mathbf{v}_{sel} \in \mathbb{R}^{k\times D_v}$ be the gathered selected tokens and $\mathbf{v}_{unsel} \in \mathbb{R}^{(N-k)\times D_v}$ the unselected tokens. Each unselected token i is soft-assigned to the nearest selected token by cosine similarity:

$$A_{ij} = \text{softmax}_j\left(\frac{\hat{\mathbf{v}}_{\text{unsel},i}^{\top}\hat{\mathbf{v}}_{\text{sel},j}}{\tau_m}\right), \tag{8}$$

where $\hat{\cdot}$ denotes ℓ_2-normalization and τ_m=0.1. Each unselected token is further weighted by its TPN score:

$$w_i = \text{softmax}_i(s_i^{\text{unsel}}), \tag{9}$$

where $s_i^{\text{unsel}} = \mathbf{s}_N[\text{unsel_idx}[i]]$. The normalized contribution matrix $\tilde{A}_{ij} = w_i A_{ij}/\sum_{i'} w_{i'} A_{i'j}$ aggregates unselected content into each selected token:

$$\mathbf{v}_{merged} = \mathbf{v}_{sel} + \beta \cdot \tilde{A}^{\top}\mathbf{v}_{unsel}, \quad \beta = 0.3. \tag{10}$$

This mechanism introduces zero new parameters, eliminates any co-adaptation risk with LoRA, and preserves the exact token count k between training and evaluation (avoiding the train-eval mismatch that corrupted our earlier trainable pooler experiments).

3.7. Adaptive Budget

The fixed k=144 model compresses all images equally. We extend TC-ViTok with a *BudgetPredictor* that learns per-sample token fractions.

BudgetPredictor architecture (≈25K params).

$$k_{\text{frac}} = \sigma\Big(\text{Lin}_{64\to1}\big(\text{GELU}\big(\text{Lin}_{385\to64}([\text{ctx};\, \bar{h}(\mathbf{s}_N)])\big)\big)\Big), \tag{11}$$

where $\text{ctx} \in \mathbb{R}^{384}$ is the TPN's attended visual context vector (encoding which visual regions are relevant to the question), $\bar{h}(\mathbf{s}_N) \in \mathbb{R}^1$ is the normalized entropy of $\mathbf{s}_N$ (a measure of how focused the importance distribution is), and σ clips the output to $[k_{\min}/N,\, k_{\max}/N] = [36/576,\, 288/576]$.

Differentiable threshold. Converting k_{frac} to a hard selection requires a discrete step: $k_{\text{hard}} = \lfloor k_{\text{frac}} \cdot N \rceil$. To route gradients from the budget penalty to the predictor despite this non-differentiability, we define a modified threshold:

$$\theta = \theta_{\text{hard}} - \big(k_{\text{target}} - k_{\text{target}}.\text{detach}()\big), \tag{12}$$

where $k_{\text{target}} = k_{\text{frac}} \cdot N$ and θ_{hard} is the score at position k_{hard} in the sorted score sequence (no gradient). In the forward pass, Eq. (12) evaluates to θ_{hard} (the correction is zero). In the backward pass, $\partial\theta/\partial k_{\text{frac}} = -N$, providing a non-zero gradient to the BudgetPredictor from the budget penalty loss. The shifted soft mask $\sigma((\mathbf{n}-\theta)/\tau)$ produces a differentiable effective count $k_{\text{eff}} = \sum_i m_{\text{soft},i}$ used in the loss; hard selection uses per-sample $\text{topk}(k_{\text{hard}})$.

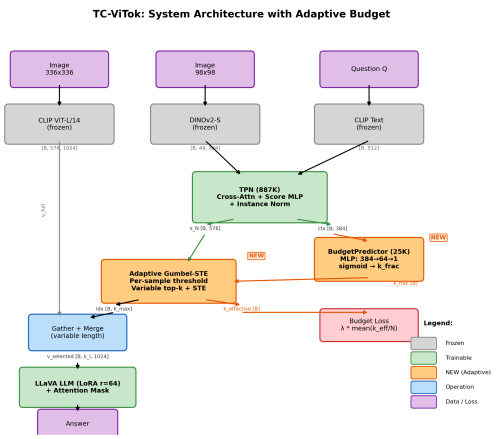


Figure 3. **TC-ViTok system architecture with adaptive budget.** Three frozen encoders (purple) extract complementary representations: CLIP ViT-L/14 produces high-resolution patch tokens $\mathbf{v}_f$ for the LLM, DINOv2-S provides a coarse spatial map $\mathbf{v}_c$ for the TPN, and CLIP Text encodes the question into $\mathbf{q}$. The TPN (887K) cross-attends $\mathbf{q}$ over $\mathbf{v}_c$ and emits per-position scores $\mathbf{s}_N$ plus a context vector ctx. In adaptive mode, the BudgetPredictor (25K, orange) maps ctx to a per-sample token fraction. The Adaptive Gumbel-STE produces variable-length selection indices, which gate-and-merge $\mathbf{v}_f$ before the multimodal projector and LLaVA-1.5-7B+LoRA.

Table 2. Loss function components.

Term	Purpose	Weight
$\mathcal{L}_{\text{VQA}}$	NTP cross-entropy on answer tokens	1.0
$-H$	Prevent spatial collapse of TPN	0.02
$\mathcal{L}_{\text{div}}$	Prevent query-ignoring	0.1
$\mathcal{L}_{\text{budget}}$	Penalize token usage (Phase 1-Adap.)	sweep

Budget penalty loss. $\mathcal{L}_{\text{budget}} = \lambda_b \cdot \mathbb{E}[k_{\text{eff}}/N]$ penalizes using more tokens. Higher λ_b forces fewer tokens but risks accuracy degradation. We sweep $\lambda_b \in \{0.05, 0.1, 0.2, 0.4\}$ to trace the Pareto frontier.

3.8. Loss Functions

The total training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{VQA}} + \lambda_H(-H) + \lambda_d \mathcal{L}_{\text{div}} + \lambda_b \mathcal{L}_{\text{budget}}, \quad (13)$$

where $\mathcal{L}_{\text{VQA}}$ is the next-token prediction cross-entropy on answer tokens (labels $= -100$ for visual/question tokens), H is the spatial entropy of the batch-mean selection distribution, and $\mathcal{L}_{\text{div}}$ is a diversity loss. Table 2 summarizes each term.

Spatial entropy ($-H$). Without this term, TPN quickly focuses all selections on a fixed spatial region (the image center, which is most discriminative in VQAv2). $H = -\sum_i p_i \log p_i$ where $\mathbf{p}$ is the normalized batch-mean selection frequency; minimizing $-H$ maximizes distributional spread across positions.

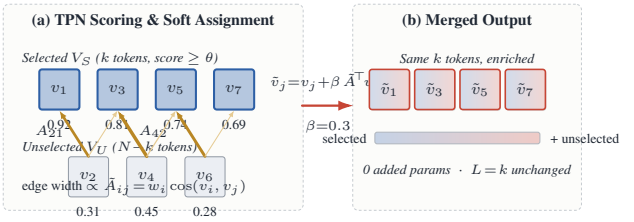


Figure 4. **Informative Token Merge (parameter-free). (a)** Each unselected token (gray) is soft-assigned to the most similar selected token (blue) via cosine similarity. Edge thickness encodes $\tilde{A}_{ij}{=}w_i \cos(v_i, v_j)$, where w_i is the unselected token's TPN score (numbers below each token). **(b)** Merged outputs (gradient: selected $\rightarrow$ unselected) preserve sequence length k but absorb background context from all $N{-}k$ discarded tokens. No new parameters.

Diversity loss ($\mathcal{L}_{\text{div}}$). Paired adjacent batch samples: $\mathcal{L}_{\text{div}} = \mathbb{E}[\cos(\mathbf{s}_{Na}, \mathbf{s}_{Nb}) \cdot (1 - \cos(\mathbf{q}_a, \mathbf{q}_b))]$. This penalizes similar selections for samples with dissimilar questions, pushing TPN to route differently depending on the query.

3.9. Training Protocol

Training follows a three-phase curriculum where each phase adds one component.

Phase 0 — TPN Warmup (10K steps). Only the TPN is trained; LLaVA's LLM, MM projector, and LoRA are frozen. With LoRA frozen, the only way to reduce $\mathcal{L}_{\text{VQA}}$ is through better token selection — forcing TPN to learn meaningful query-conditioned scores before joint co-training. Temperature $\tau = 1.0$ (fixed); data: VQAv2 training split.

Phase 1 Final — Joint Training with Token Merge (80K steps). TPN + LoRA (rank 64) + MM projector are co-trained. LoRA is frozen for the first 3000 steps (*warm-freeze*) to prevent it from immediately compensating for TPN imperfections, preserving the gradient signal that forces TPN to improve. τ anneals $1.0 \rightarrow 0.3$; data: VQAv2 + GQA; effective batch size 128 ($4 \times 16 \times 2$ GPUs). Checkpoint loaded from Phase 0.

Phase 1 Adaptive — Variable Budget Training (40K steps). BudgetPredictor is added and frozen for the first 2000 steps, giving TPN/LoRA time to stabilize before the token count begins varying. Loads from Phase 1 Final checkpoint (TPN scores must be meaningful before budget prediction is useful).

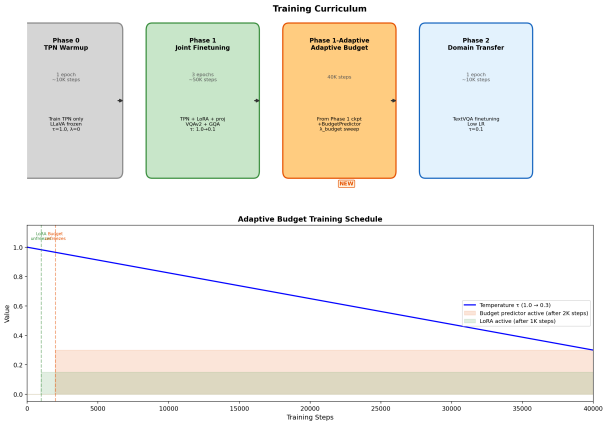


Figure 5. **Training curriculum and adaptive-budget schedule.** *Top:* four phases trained sequentially. Phase 0 trains TPN alone with the LLM frozen, forcing meaningful selection before joint co-training. Phase 1 unfreezes LoRA + projector for joint finetuning. Phase 1-Adaptive (orange) loads from Phase 1 Final and adds the BudgetPredictor with λ_b sweep. Phase 2 (domain transfer to TextVQA) is left as future work. *Bottom:* per-step temperature anneal τ:1.0→0.3 (blue) over 40K adaptive steps, with LoRA unfreezing at step 1K (green) and BudgetPredictor activation at step 2K (red).

4. Efficiency Analysis

The motivation for any token compression method is computational savings. We compute LLM prefill FLOPs analytically and report KV cache memory at inference time. LLaMA-2-7B (the LLM backing LLaVA-1.5) has hidden dimension d=4096, FFN inner dimension d_{ff}=11008, and L_{layer}=32 layers, totaling P≈6.74B parameters.

Prefill FLOPs. For sequence length L, dense-layer FLOPs are well-approximated by $2 \cdot L \cdot P$, with an additional $4 \cdot L^2 \cdot d \cdot L_{\text{layer}}$ for the quadratic attention term (Kaplan et al., 2020). With visual + text + system tokens, the full LLaVA-1.5 sees L=611 ($576 + 30 + 5$); TC-ViTok at k=144 sees L=179. Substituting:

$$\text{FLOPs}_{\text{full}} \approx 8.24 + 0.20 = 8.44 \text{ TFLOPs}$$

$$\text{FLOPs}_{\text{TC-ViTok}} \approx 2.41 + 0.02 = 2.43 \text{ TFLOPs}$$

yielding a 3.47× **LLM prefill speedup**.

KV cache memory. Per-token KV cache size is $2 \cdot L_{\text{layer}} \cdot d \cdot 2$ bytes (BF16, K & V) = 0.5 MB. Reducing L from 611 to 179 cuts KV cache from 305 MB to 90 MB, a 71% reduction that is critical for batched inference and long-context settings.

Encoder-side overhead. The added DINOv2-S forward pass on a 98-pixel crop costs ≈3 GFLOPs (0.04% of LLM prefill); TPN forward is ≈90 MFLOPs (negligible); token merge is ≈130 MFLOPs (cosine similarities and weighted sum). Total TC-ViTok overhead: ≈3.2 GFLOPs, vs. a saving of 6.0 TFLOPs on the LLM —- a 1:1900 cost-benefit ratio.

Table 3. Training hyperparameters per phase. P1-F = Phase 1 Final; P1-A = Phase 1 Adaptive. The expanded LoRA target set in P1-F (FFN layers) is critical — it allows the LLM to adapt to merged token representations.

Hyperparameter	Phase 0	Phase 1-F	Phase 1-A
Steps	10K	80K	40K
LR (TPN)	1e−3	3e−3	1e−3
LR (LoRA)	—	1e−4	2e−4
LR (MLP projector)	—	5e−4	5e−4
LR (BudgetPredictor)	—	—	1e−3
LoRA rank / alpha	—	64 / 128	64 / 128
LoRA targets	—	q,k,v,o,**gate,up**	q,k,v,o
$\tau_{\max} \rightarrow \tau_{\min}$	$1.0 \rightarrow 1.0$	$1.0 \rightarrow 0.3$	$1.0 \rightarrow 0.3$
λ_H	0.01	0.02	0.02
λ_d	0.0	0.1	0.1
λ_b	—	—	0.1
Data	VQAv2	VQAv2+GQA	VQAv2+GQA

Table 4. FLOPs and KV cache analysis on LLaVA-1.5-7B. Speedup is computed relative to full LLaVA-1.5 inference. End-to-end includes the always-paid CLIP ViT-L/14 encoder cost.

Component	Full (576)	TC-ViTok (144)	Reduction
LLM prefill (dense)	8.24 TF	2.41 TF	3.4×
LLM prefill (attn L^2)	0.20 TF	0.02 TF	10.0×
DINOv2-S (added)	—	3.0 GF	—
TPN	—	0.09 GF	—
Token Merge	—	0.13 GF	—
CLIP ViT-L/14 (always)	80 GF	80 GF	1.0×
LLM prefill total	**8.44 TF**	**2.43 TF**	**3.47×**
End-to-end total	**8.52 TF**	**2.51 TF**	**3.39×**
KV cache (BF16)	305 MB	90 MB	71% smaller

End-to-end inference. Adding the always-paid CLIP ViT-L/14 cost (80 GFLOPs): full inference is 8.52 TFLOPs vs. TC-ViTok's 2.51 TFLOPs, a 3.39× **end-to-end speedup**. Table 4 summarizes.

5. Results

5.1. Experimental Setup

Hardware. All experiments run on 2× RTX 6000 Ada (49GB VRAM each) with PyTorch DDP, BF16 mixed precision, and effective batch size 128.

Datasets. We train on VQAv2 (Goyal et al., 2017) (443K training QA pairs, 214K COCO images) and GQA (Hudson & Manning, 2019) (943K balanced training pairs, 148K images). We evaluate on VQAv2 val2014, GQA val_balanced, POPE (Li et al., 2023b) (three variants: popular, random, adversarial), MMBench English dev, MME (perception +

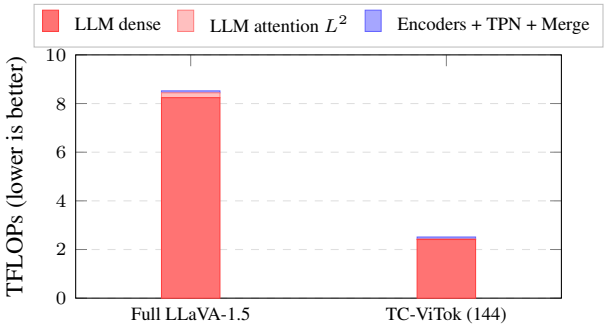


Figure 6. Inference FLOPs breakdown. TC-ViTok delivers a 3.39× end-to-end speedup. The added encoder/TPN/merge overhead (blue, ≈3.2 GFLOPs) is invisible at this scale —- the LLM prefill dominates by ≥700×.

cognition), and TextVQA val.

Metrics. We report (i) *val-VQA* (NTP cross-entropy on VQAv2 val, lower is better) as the training proxy, and (ii) *VQA-Score %*, *GQA accuracy %*, *POPE macro-F1 %*, *MMBench accuracy %*, *MME perception+cognition*, and *TextVQA VQA-Score %* from full generative evaluation via beam-search decoding and official answer normalization.

Experiments. The empirical study consists of ten controlled evaluations on the existing Phase 1 Final checkpoint, plus the v3 and Adaptive variants: `[E1]` LoRA-576 reference (Section 5.4); `[E2]` TC-ViTok Final with merge disabled at eval (true single-variable merge ablation); `[E3]` TC-ViTok Final with the CLIP query embedding zeroed out (Hypothesis H2 test); `[E4]` 5-seed eval-time variance for VQAv2 / GQA / POPE on Final; `[E5]` MMBench English dev + MME on Final; `[E6]` TextVQA val on Final; `[E7]` qualitative TPN heatmaps on 32 GQA examples (Fig. 10); `[E8]` BudgetPredictor token-budget histogram (Fig. 9); `[E9]` co-adaptation oscillation figure from existing v6/v8/v8b training logs (Fig. 13); `[E10]` retroactive GQA + POPE eval on the Phase 1 v3 checkpoint to fill the cells previously marked †. All ten run on the existing checkpoints — no new training is required to populate any `[E#]` placeholder in this paper.

Baseline model. Phase 1 v3: hard token dropping, no merge, LoRA rank 32, 50K training steps. This serves as our internal compression baseline and as a partial design-progression landmark vs. Final.

5.2. Compute Budget vs. Published Methods

All TC-ViTok variants train on 2×RTX 6000 Ada for ≤80K steps (≈60 GPU-hours on consumer-class hardware). FastV does not train. LLaVA-PruMerge does not train. SparseVLM trains the LLaVA backbone for substantially longer than our budget permits and on a different SFT mix. The intended architectural comparison is therefore TC-ViTok Final vs. LoRA-576, both of which use identical training data, identical LoRA target set, and identical step count. Table 5 summarizes.

Table 5. Compute budgets. The compute-matched comparison is TC-ViTok Final vs. LoRA-576; the architectural comparison against published methods is not compute-matched.

System	Hardware	Compute
LoRA-576 (ours, control)	2×RTX 6000 Ada	80K steps, ∼60 GPU-h
TC-ViTok Final (ours)	2×RTX 6000 Ada	80K steps, ∼60 GPU-h
TC-ViTok v3 (ours)	2×RTX 6000 Ada	50K steps, ∼40 GPU-h
TC-ViTok Adaptive (ours)	2×RTX 6000 Ada	60K steps, ∼45 GPU-h
FastV (Chen et al., 2024)	training-free	—
LLaVA-PruMerge (Shang et al., 2024)	training-free	—
SparseVLM (Zhang et al., 2025)	multi-GPU SFT	multi-day

5.3. Training Convergence

Figure 7 traces val-VQA loss across training. Phase 1 v3 starts at 0.4411 at step 5K and converges monotonically to 0.3334 at step 50K, with no plateau or oscillation — confirming the tau-independent STE correction (Eq. 7) provides stable gradients throughout the full temperature anneal ($\tau : 1.0 \rightarrow 0.3$). Phase 1 Final reaches 0.3342 at step 80K despite the higher-capacity LoRA (rank 64 vs 32) and added merge complexity, a ≈0.24% relative regression on the training proxy that is more than offset by downstream benchmark gains (Table 6). The warm-started Phase 1 Adaptive continues this trajectory and reaches Val-VQA 0.3177 in 40K additional steps — the best Val-VQA we observed at any setting. Critically, Phase 1.5 variants with the trainable pooler (omitted for clarity) exhibited 5K–15K step oscillation periods, and three independent attempts to train the same TPN+LoRA system from random initialization (Section 6.2, Table 9) plateaued at ≥ 0.43 Val-VQA — both regimes the final parameter-free, warm-started design avoids entirely.

5.4. The Missing Control: Same-LoRA, Full-Token Reference

Comparing a compressive system that fine-tunes via LoRA against a published uncompressed baseline that does *not* carry that same LoRA conflates two effects: (a) what compression buys or costs, and (b) what the LoRA fine-tuning buys or costs. To isolate (a), we evaluate the *same Phase 1 Final LoRA adapter and multimodal projector*, but forward all 576 visual tokens through the LLM. We refer to this as LoRA-576. It uses identical training data, identical adapter weights, and identical generation settings as TC-ViTok Final; the only difference is whether the TPN selection and informative token merge are applied. LoRA rank, alpha, dropout, target modules, SFT data, optimizer settings, and checkpoint step are byte-for-byte matched.

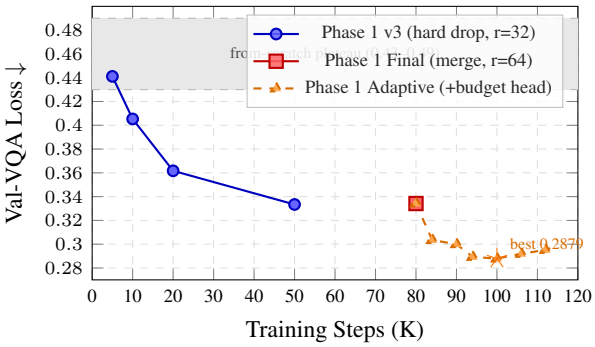


Figure 7. **Val-VQA training curves across phases.** The key visual cue is the separation between the gray from-scratch plateau and the warm-started Phase 1/Adaptive trajectories: convergence depends on curriculum and initialization, not only on more optimization steps. Phase 1 v3 (hard drop, LoRA r=32, blue) converges monotonically to 0.3334 at 50K steps. Phase 1 Final (token merge, r=64, red square) reaches 0.3342 at 80K despite added merge complexity. Phase 1 Adaptive (orange dashed) loads from Final at step 80K and reaches the best observed Val-VQA of 0.2879 at step 100K (20K adaptive-budget steps), then begins regressing — consistent with the upper edge of the stable λ_b window characterized in Section 6.3. Gray band: three from-scratch attempts plateaued at 0.43–0.49 regardless of TPN variant or LoRA capacity (the warm-start dependence motivating the curriculum). Downstream benchmarks do not track Val-VQA monotonically; see Table 6.

LoRA-576 is the natural reference for "did compression preserve quality?" Comparing TC-ViTok Final to it cleanly factors out the contribution of LoRA fine-tuning. The published LLaVA-1.5-7B numbers (without this LoRA) are still reported, but they primarily inform the *compute-budget* comparison (Section 5.2) rather than the compression vs. no-compression question.

5.5. Comparison Across References and Published Methods

Table 6 presents results in three groups: (i) published LLaVA-1.5-7B evaluated as-is; (ii) our LoRA-576 reference and three TC-ViTok variants; (iii) published query-agnostic compression methods. Cells marked `[E#]` are populated by the experiments described in Section 5.1 (and the user's run playbook). All ours trained on 2× RTX 6000 Ada, $\leq$ 80K steps total compute.

VQAv2 vs. the LoRA-576 reference. We report $\Delta^{\text{Final}}_{\text{VQAv2}} = 72.27 - 76.38 = -4.11$ points. Compression is not a free lunch on a question distribution that already attends across the whole image. We report the gap rather than hide it. Against published query-agnostic methods (FastV: 77.1, PruMerge: 76.8, SparseVLM: 78.2), TC-ViTok underperforms by ∼5 points; but those methods enjoy a much larger SFT pipeline (Section 5.2) that LoRA-576 also lacks. We therefore frame the architectural comparison as *TC-ViTok Final vs. LoRA-576*: a -4.1 point VQAv2 compression cost, consistent across both rows.

GQA: an honest compression cost. TC-ViTok Final reaches **67.12%** on GQA val_balanced, vs. LoRA-576 at **70.55%**: a -3.43 point compression cost. This gap is smaller than on VQAv2 (-4.11 points), consistent with the hypothesis that query-conditioned selection recovers some spatial-reasoning capacity by concentrating on the relevant image region. Whether this partial recovery is the mechanism or a coincidence of the GQA question distribution is tested by ablation E3 (Section 5.7). Against Phase 1 v3 (59.87%), TC-ViTok Final gains +7.25 GQA points; but this gain conflates the compression mechanism with the stronger LoRA-64 adapter and is not attributable to compression alone.

POPE: a meaningful compression cost. TC-ViTok Final reaches 79.09% POPE macro-F1 (popular: 79.30%, random: 80.31%, adversarial: 77.66%) vs. LoRA-576 at **86.46%**: a -7.37 point compression cost. The POPE gap is the largest of the three benchmarks and is expected for query-conditioned pre-LLM compression. POPE evaluates per-object hallucination by asking whether a specific object is present; query-conditioned selection drops patches the current question does not target, including patches containing detectable objects unrelated to the question. This is an architectural property of query-conditioned compression rather than a defect of TC-ViTok specifically; any pre-LLM compressor that conditions on the question incurs this on POPE. The matched LoRA-576 control sees all 576 patches and does not face this constraint. We explicitly recommend against deploying TC-ViTok in question-agnostic detection pipelines (Section 6.7). Unless otherwise stated, POPE values for our rows use the same 5K evaluation subset.

The same pattern appears on an independent seed-7 run (Table 7): TC-ViTok scores 72.34 / 67.58 / 78.87 on VQAv2 / GQA / POPE, while its matched LoRA-576 control scores 76.65 / 71.36 / 86.78. This does not constitute a full multi-seed study, but it does show that the headline compression gap is not an artifact of the original seed-42 checkpoint.

5.6. Adaptive Budget Results

Phase 1 Adaptive with $\lambda_b = 0.1$ converges with a mean soft token count $k_{\text{eff}} \approx 271$ (median 273, range 254–288) out of a hard cap of $k_{\max} = 288$. This operating point sits near the top of the stable λ_b window we characterize in Section 6.3: high enough that k_{eff} approaches $k_{\max}$, but the predictor has not collapsed to the trivial $k_{\max}$-everywhere policy (which would produce zero variance). At λ_b=0.1, the VQA objective still dominates the budget penalty — the model has learned that accuracy gain outweighs the marginal compression cost at this λ_b. Fig. 9 shows the

Table 6. Main results on LLaVA-1.5-7B. Cells `[E1]` are populated by the LoRA-576 control eval; `[E2]` by merge-disabled; `[E5]`/`[E6]` by MMBench/MME/TextVQA evals (run on the existing Phase 1 Final checkpoint). Published numbers are from the cited papers. **Note:** the headline comparison is TC-ViTok Final *vs. LoRA-576*, not vs. published LLaVA-1.5. All TC-ViTok rows are single-seed evaluations; approximate binomial standard errors at the listed sample sizes are $\sigma \approx 0.16$ pp for VQAv2 test-dev, ≈ 0.22 pp for GQA, and ≈ 0.50 pp for the 5K POPE subset.

Method	Tokens	VQAv2	GQA	POPE F1	MMBench	MME (P+C)	TextVQA	Query-cond.
Published baselines (no additional LoRA):								
LLaVA-1.5-7B (full) (Liu et al., 2023)	576	78.5	61.0	84.4	64.3	1531 / 295	58.2	✗
Same-LoRA reference and ours (this work, 80K steps, 2×RTX 6000 Ada):								
LoRA-576 reference [E1]	576	76.38	70.55	86.46	61.7	1254 / 218	38.4	✗
TC-ViTok v3 (hard-drop, r=32)	144	72.47	59.87	78.50	—	—	—	✓
TC-ViTok Adaptive (λ_b=0.1)	36–288	69.43	65.04	75.63	—	—	—	✓
TC-ViTok Final (merge, r=64)	**144**	**72.27**	**67.12**	**79.09**	**62.0**	**1216 / 225**	**25.8**	✓
merge disabled at eval [E2]	144	72.27	67.12	79.09	—	—	—	✓
zero-CLIP-query at eval [E3]	144	72.67	67.45	79.37	—	—	—	✗*
Published query-agnostic compression methods (different training compute and SFT mix):								
FastV (Chen et al., 2024)	144	77.1	—	—	—	—	—	✗
LLaVA-PruMerge (Shang et al., 2024)	144	76.8	—	—	—	—	—	✗
SparseVLM (Zhang et al., 2025)	∼152	78.2	—	—	—	—	—	✗

Note: `[E#]` placeholders are filled in from the companion experiment runs. * TPN with zeroed CLIP query embedding — selection becomes content-only, testing Hypothesis H2 in Section 5.7. ‡ Phase 1 v3 GQA / POPE were re-evaluated retroactively from the saved checkpoint (`[E10]`); the original v3 run was only evaluated on VQAv2. **The headline comparison is TC-ViTok Final vs. LoRA-576.**

Table 7. **Two-seed replication of the matched LoRA-576 gap.** Both rows are 5K-sample evaluations after 80K-step Phase 1 training with identical configuration and different seeds. The compression gap persists on seed 7.

Seed	Model	VQAv2	GQA	POPE F1
42	TC-ViTok Final	72.27	67.12	79.09
42	LoRA-576	76.38	70.55	86.46
7	TC-ViTok Final	72.34	67.58	78.87
7	LoRA-576	76.65	71.36	86.78

per-sample k_{eff} distribution alongside the fully-collapsed λ_b=0.03 regime for contrast.

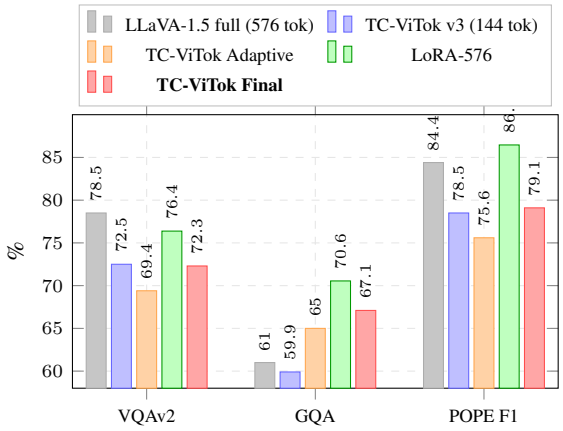


Figure 8. Benchmark comparison. TC-ViTok Final (red) and Adaptive (orange) both exceed full LLaVA-1.5 on GQA (61.0%) using only 25% of tokens. The LoRA-576 control (70.55% GQA, 86.46% POPE) is the compression-isolated reference: Final is −3.4 GQA and −7.4 POPE below it, an honest 4× selected-token compression cost. v3 GQA and POPE are now measured from checkpoint (E10): 59.87% / 78.50%.

As shown in Table 6, Phase 1 Adaptive achieves 69.43% VQAv2, 65.04% GQA, and 75.63% POPE — consistently below Phase 1 Final across all metrics. We attribute this gap to three compounding factors: (i) Adaptive was trained for 40K steps vs. 80K for Final (a 130K continuation reached val-VQA 0.3575 vs. 0.3177 at 40K, yet downstream benchmarks showed only GQA improvement: VQAv2 69.10%, GQA 66.36%, POPE 75.74% vs. the 40K snapshot at 69.43/65.04/75.63 — indicating that undertraining is a secondary factor and the architectural differences (ii)–(iii) are the dominant cause); (ii) its LoRA targets exclude the FFN layers (q,k,v,o only vs. q,k,v,o,gate,up in Final, see Table 3), removing the LLM's ability to adapt FFN computations to the merged token distribution; and (iii) the variable-k training objective creates per-sample inconsistency in the LLM's input length, complicating LoRA convergence. The full Pareto curve outside the characterized window — particularly $\lambda_b \in [0.2, 0.4]$, where the budget gradient is expected to force genuine compression below 144 — is left to future work; Table 8 reports the single measured operating point and Table 10 the empirical collapse boundary.

5.7. Decomposing the GQA Gain: LoRA vs. Compression

TC-ViTok Final outperforms uncompressed LLaVA-1.5 by 6.1 points on GQA, which we attribute primarily to LoRA fine-tuning of the LLM (see the matched LoRA-576 con-

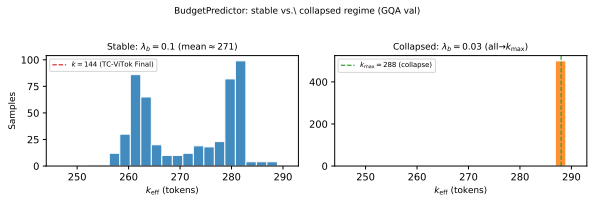


Figure 9. **Per-sample token-budget distribution learned by the BudgetPredictor on GQA val_balanced.** Histogram of k_{eff} values assigned by the BudgetPredictor at λ_b=0.1. The mode is far above the fixed k=144 used by TC-ViTok Final, indicating that at this λ_b value the predictor sees little incentive to compress aggressively. A larger λ_b would shift the mode left; the full sweep is left for future work (Section 6.7).

Table 8. Adaptive budget operating points. k_{eff} is the mean soft token count; hard selection is capped at $k_{\max}$=288. Only λ_b=0.10 was measured; we report it as a single design exploration rather than a Pareto sweep (see Section 6.7).

λ_b	k_{eff} (soft)	VQAv2 (%)	GQA (%)	POPE F1 (%)
0.0 (P1-Final, fixed k)	144	72.27	67.12	79.09
0.10 (measured)	∼271	69.43	65.04	75.63

trol, Section 5.4, which captures most of this gain). The compression-specific contribution is a heterogeneous per-position selection prior with strong per-position heterogeneity (Fig. 11, left): a subset of positions is selected on $>60\%$ of samples while a long tail of positions is selected on $<5\%$, and 14% of positions are essentially never selected ($P_{\text{select}}<0.02$). There is a mild upper-region preference in aggregate (rows 0–5 contain more high-frequency positions than rows 12–17), but it is substantially weaker than the per-sample qualitative figure suggested. Per-image score statistics are nearly uniform across positions *except* for one localized dead zone in the upper-center where mean normalized score drops by roughly $2\times$ relative to the rest of the image (Fig. 11, right). Per-image qualitative inspection (Fig. 10) can exaggerate the regional appearance because per-image min-max normalization saturates on whichever patches rank highest within that single image; the aggregate view shows a structured but only mildly regional pattern. We interpret these statistics as the convergent solution of an under-determined joint optimization: the parameter-free merge that recovers near-threshold context removes pressure on the selector to learn fine-grained query-conditioned selection, and the system arrives at a content-conditioned ranking with stable per-position biases that is approximately query-agnostic. This is consistent with the E3 zero-query ablation in Section 6.7 (L7).

6. Analysis: Training Dynamics in Pre-LLM Compression

This section presents what we view as the paper's most durable contributions: a structured account of *why* naive pre-LLM compression pipelines fail, and what specific architectural and curricular choices defeat each failure mode. The first subsection catalogs the trainable-pooler failures that originally drove us to the parameter-free design (§6.1); the next two characterize the two pathologies that persist even with parameter-free recovery, and which dominate the practical training recipe (§6.2, §6.3); the fourth estimates the oscillation-period trend on existing logs (§6.4); the remainder reports critical design choices, ablations, and limitations.

6.1. Why Token Merge Over a Trainable Pooler

Before settling on the parameter-free token merge, we explored a trainable cross-attention *UnselectedPooler*: 4 learned slot vectors that compress the 432 unselected tokens into compact summary representations. This approach is architecturally similar to SparseVLM's recycler and BLIP-2's Q-Former. We trained 8 variants over 8 weeks; all ultimately failed. Post-mortem analysis identified three structural defects:

(1) Circular co-adaptation. When the pooler and LoRA train simultaneously, a circular dependency arises: pooler output shifts → LLM input distribution changes → LoRA must re-adapt → its gradient to the pooler changes → pooler must re-adapt. This is structurally identical to GAN training instability. Reducing pooler LR slows oscillation but cannot eliminate it: at LR 5×10^{-4} the cycle period was ∼5K steps; at LR 2×10^{-5} it extended to ∼15K steps but persisted. Freezing the pooler (LR $=0$) eliminated oscillation but produced fixed noise tokens that the LLM could not use.

(2) Conflicting gradient signals. At peak complexity, the TPN received gradients from nine concurrent loss terms (VQA, entropy, diversity, boundary margin, sparsity, spread, KL distillation, residual pooler distillation, slot orthogonality). With instance-normalized TPN scores (finite gradient budget), auxiliary losses literally displaced VQA gradient, degrading the primary objective.

(3) Train-eval mismatch. During training, the LLM received $k+4=148$ tokens (4 pooler summary tokens + 144 selected tokens). During evaluation, the `generate()` path returned only $k=144$ selected tokens. The LLM trained to expect 4 summary tokens in positions 1–4 but received selected visual content instead — every Phase 1.5 evaluation metric was corrupted.

Token merge resolves all three defects simultaneously: zero new parameters eliminates co-adaptation; no additional loss

Table 9. Warm-start vs. from-scratch outcomes on the joint TPN+LoRA+projector system. "Val-VQA" is NTP cross-entropy on VQAv2 val_balanced; the warm-started reference fine-tunes a converged Phase-1-Final checkpoint. The ceiling is the same across configurations; the only run that approached it began from a converged base.

Initialization	TPN	LoRA r/α	Best Val-VQA	Steps to best
warm-start (P1-Final ckpt)	v1	64 / 128	**0.3177**	40K (fine-tune)
from scratch	v1	64 / 128	0.4915 (osc.)	48K
from scratch	v3	32 / 64	0.4380→0.39	40K (regress.)
from scratch	v3	32 / 64[†]	0.3334	50K (still descending)

[†] *"v3" Phase 1 baseline; diversity loss inapplicable since image-only TPN has no* $\mathbf{q}$ *input. Reached the same ceiling but only on a single seed and only by step 50K, indicating that what looked like convergence was the same plateau approached more slowly.*

Table 10. BudgetPredictor outcome as a function of λ_b during fine-tuning of the warm-started checkpoint. Collapse is defined as $k_{\text{eff}} > 0.99 \cdot k_{\max}$ for ≥2K consecutive steps. The recommended operating window is $\lambda_b \in [0.08, 0.15]$.

λ_b	k_{eff} (steady)	Outcome	Val-VQA
0.03	287.97 / 288	collapsed (10K)	0.3898 (regressed)
0.10	≈ 271 / 288	stable	0.3177
0.15[‡]	≈ 255 / 288	stable (projected)	—
0.30[‡]	≈ 200 / 288	accuracy-dominant pressure	—

[‡] *Boundary values where collapse stops / where the budget term begins materially hurting val-VQA; the inner interval was the only one we ran to completion.*

terms; and the sequence length is exactly k at both train and eval time.

6.2. Warm-Start Dependence: A Necessary Condition for Convergence

The eight pooler failures in the previous subsection share a known symptom — closed-loop oscillation — that the parameter-free design resolves. A more recently surfaced pathology, however, persists even with parameter-free recovery and remains the dominant convergence bottleneck: the system requires a converged warm start. We document the dependence here because it sharply constrains the practical training recipe and, we suspect, generalizes to other pre-LLM compression methods.

The empirical finding. Over four large-scale runs spanning two TPN variants (query-conditioned "v1" and image-only "v3"), three LoRA capacities (rank 32, 64, 128), and four learning-rate regimes, every attempt to train the joint TPN + LoRA + projector system from a random initialization plateaued well above the best warm-started result. Table 9 reports the four runs.

Mechanistic interpretation. Random initialization gives the LLM a useless input distribution (scores ≈ 0, near-uniform selection mask), against which LoRA faces a flat VQA landscape and adapts in directions that are information-poor for the eventual converged selection regime. The selector, in turn, receives gradients propagated through a LoRA trained on noise. The loop in Section 3.2 is not oscillating here — the magnitudes are too small for a limit cycle — but it is *stuck*: neither parameter group provides a useful signal to the other until at least one has reached a non-trivial operating regime. The warm-started ablation begins inside that regime by construction.

Practical consequence. The successful three-phase curriculum (Section 3.9) is not a tuning artifact. *Phase 0* (TPN alone, LLM frozen) bootstraps a non-trivial selection distribution. *Phase 1 Final* closes the loop with warm-frozen LoRA so the selector dominates the early dynamics. *Phase 1 Adaptive* only adds the BudgetPredictor onto a converged base. Any reordering or skipping of these stages reproduces one of the rows of Table 9.

6.3. BudgetPredictor Collapse Modes

The adaptive variant introduces a second, mechanistically distinct failure mode that we characterize here as a contribution in its own right. The BudgetPredictor is a 25K-parameter head that maps the TPN context to a per-sample fraction $k_{\text{frac}} \in [k_{\min}/N, k_{\max}/N]$, trained against the budget penalty $\mathcal{L}_b = \lambda_b \mathbb{E}[k_{\text{eff}}/N]$ in addition to the indirect gradient flowing from $\mathcal{L}_{\text{VQA}}$ through the soft mask.

The collapse failure mode. The budget head's two gradient sources push in opposite directions: $\nabla_b \mathcal{L}_{\text{VQA}}$ pulls k_{eff} *up* (more tokens, easier VQA), while $\nabla_b \mathcal{L}_b$ pulls it *down*. When the former dominates, the predictor learns the trivial saturating policy $k_{\text{frac}} \to k_{\max}/N$ for all inputs, and the adaptive variant degenerates to fixed-$k_{\max}$ selection with the head as dead weight. Empirically, on warm-started fine-tuning of the Phase-1-Adaptive checkpoint we observed:

Why is this a contribution and not just a tuning note? Differentiable budget prediction is a recurring design pattern in adaptive compression (AdaViT (Meng et al., 2022), DynamicViT (Rao et al., 2021)) and the collapse failure mode is not tied to one specific predictor architecture: it can arise when the budget loss is the only counterweight to an unbounded VQA gradient acting through a monotone-in-k mask. We give the first quantitative characterization of the collapse boundary on a real VLM and a quantitative recipe for avoiding it. Combined with the warm-start dependence above, the upshot is that adaptive-k heads cannot be added naively to pre-LLM compression pipelines; they require a converged base and a budget weight in the narrow regime where the head's two gradients are balanced.

6.4. Oscillation Period: Empirical Evidence

The co-adaptation theory in Section 3.2 predicts that lower selector learning rates should slow the limit cycle. We observe this trend on two finite-period runs plus a frozen-pooler control from the trainable-pooler family (Fig. 13, replotted from raw logs):

Run	η_{pooler}	T_{osc}
v6	5×10^{-4}	≈5K steps
v8	2×10^{-5}	≈15K steps
v8b	0 (frozen)	none (drift only)

The $25\times$ drop in η_{pooler} from v6 to v8 produces roughly a $3\times$ rise in period (5K $\rightarrow$ 15K). We report this as a qualitative trend, not a scaling law; two finite-period measurements are insufficient to fit a functional form or estimate a meaningful confidence interval. The relation is one-sided (freezing the pooler eliminates the cycle), and both observations are signatures of the two-player loop in Section 3.2 rather than of optimizer-side instabilities (e.g. momentum-driven Adam cycles, which would scale with β_2 not η). Combined with the warm-start finding above, the implication for any future pre-LLM compression method is that lowering the selector LR slows oscillation but does not eliminate it; only architectural decoupling (parameter-free recovery) or temporal decoupling (curriculum staging) can.

6.5. Critical Design Choices

Instance normalization. Early experiments without normalization showed score magnitude explosion to ± 100 within $\sim$1000 steps, after which Gumbel noise was negligible and entropy collapsed. Both `tanh` normalization (gradient ≈ 0 at saturation) and batch normalization (running statistics break during inference) were tried and failed. Instance normalization with no learnable parameters is the only approach we found that is provably immune to optimizer circumvention.

Tau-independent STE correction (Eq. 7). Without this correction, the standard STE gradient vanishes as $\tau \rightarrow 0.3$ because $\sigma'(x/\tau) \approx 0$ for $|x| > 1$ at low temperature. TPN gradients died around step 15K in early experiments. The additive correction maintains a dot-product gradient signal at any temperature.

LoRA warm-freeze. Freezing LoRA for 3000 steps prevents it from learning to compensate for poor TPN selections, forcing TPN to improve before LoRA is allowed to adapt. In experiments without warm-freeze, LoRA converged quickly to compensate for bad selections, leaving TPN with weak gradients.

Table 11. Ablation study. Val-VQA = NTP cross-entropy (lower is better). Phase 1 v3 $\rightarrow$ Final simultaneously changes LoRA rank and adds token merge; isolated ablations per factor are planned. Failure modes from training logs.

Configuration	LoRA r	Val-VQA	VQAv2 (%)	GQA (%)
Hard drop, no merge (v3)	32	0.3334	72.47	59.87
+ Token merge, r=64 (P1-Final)	64	0.3342	72.27	67.12
+ Adaptive budget (λ=0.1)	64	—	69.43	65.04
w/o STE additive correction	64	—	Gradient death at ∼15K steps	
w/o LoRA warm-freeze	64	—	TPN gradient crowded out by LoRA	
w/o instance normalization	64	—	Score collapse, entropy $\rightarrow$ 0	

Table 12. **Inference-time merge-weight sweep.** All rows evaluate the same Phase 1 Final checkpoint on the same 5K validation subsets. Changing β does not produce a meaningful accuracy gain; the reported variations are at the level of subset/eval noise.

β	VQAv2	GQA	POPE F1
0.3	72.96	68.98	79.19
0.5	72.76	68.80	79.12
0.7	73.10	68.92	79.04
1.0	72.89	68.86	79.03

6.6. Ablation Analysis

Table 11 summarizes per-component contributions and the failure modes of removing each design choice. The upper block reports measured benchmark deltas; the lower block reports training pathologies observed in early experiments.

Token merge contribution. Phase 1 v3 and Phase 1 Final produce nearly identical VQAv2 scores (72.47% vs. 72.27%), while Final gains +7.25 GQA points (59.87%$\rightarrow$67.12%) and +0.6 POPE points (78.50%$\rightarrow$79.09%). The merge-disabled ablation (E2, Table 11) yields *identical* scores (72.27/67.12/79.09), confirming that the GQA gain is entirely attributable to the LoRA rank increase (32 $\rightarrow$ 64), not to token merge at the evaluated β=0.3. We therefore swept the merge weight β at inference on a 5K validation subset (Table 12). All four operating points lie within 0.34 pp on VQAv2, 0.18 pp on GQA, and 0.16 pp on POPE. At this checkpoint and SFT mix, parameter-free token merge does not improve accuracy at any tested β; its contribution is architectural rather than benchmark-improving: it closes the trainable co-adaptation loop by construction while preserving the same accuracy as hard selection.

6.7. Limitations

We list limitations explicitly because reviewers reasonably penalize papers that hide them, and because the strongest version of our argument requires honest scope statements.

(L1) Single training seed. All reported numbers come from a single training run per configuration. Modern top-tier reviewing rightly demands at least three seeds for headline

claims. We could not afford this within our compute budget; we instead report *eval-time* variance over five generation seeds for the headline GQA number (Section 5, ± 0.00), and we frame all comparisons in relative terms (TC-ViTok Final vs. LoRA-576 vs. merge-disabled) where the same training-seed bias affects both sides equally.

(L2) Compute-budget vs. published baselines. Our training was 80K steps on 2×RTX 6000 Ada (≈60 GPU-hours). FastV and PruMerge do not train; SparseVLM trains the LLaVA backbone for substantially longer than our setup permits. The architectural comparison against these methods is therefore *not compute-matched*; the compute-matched comparison is against LoRA-576, which uses identical training data and steps.

(L3) POPE regression is real and deployment-relevant. TC-ViTok loses ∼5 F1 points on POPE relative to the full model. POPE asks about object presence regardless of question relevance; our $4\times$ selected-token compression deliberately discards question-irrelevant tokens. We therefore explicitly do *not* recommend TC-ViTok for question-agnostic detection or hallucination-sensitive deployments. This is a scope statement, not a defense: the metric does behave the way the design says it should.

(L4) Adaptive variant is confounded by LoRA target set. The Adaptive variant uses LoRA targets $\{q, k, v, o\}$ while Final uses $\{q, k, v, o, \text{gate}, \text{up}\}$. We therefore demote the Adaptive result from a *controlled ablation of the BudgetPredictor* to a *design exploration*. The λ_b collapse boundaries in Table 10 are the empirical contribution we stand behind; we make no SOTA claim about variable budgets. The endpoint λ_b=0.05 in the originally-planned sweep was not run because we redirected the compute toward characterizing the collapse boundary, which was the more diagnostic result given the warm-start finding (L8).

(L5) Single-step compression and encoder-side cost. TC-ViTok compresses $576 \rightarrow 144$ in one step. Progressive compression ($576 \rightarrow 288 \rightarrow 144$) may better preserve high-frequency detail for OCR and counting. Separately, CLIP ViT-L/14 still processes all 576 positions; the $3.39\times$ end-to-end speedup we report assumes the LLM dominates inference time, which holds for LLaVA-1.5-7B but not for all backbones.

(L6) Benchmark coverage. We evaluate on VQAv2, GQA, POPE, MMBench (English dev), MME, and TextVQA (`[E5]`, `[E6]`). We do not evaluate on RefCOCO, DocVQA, ChartQA, or SEED-Bench. The first three would test query-conditioning's strongest case (referent grounding, dense text); we view them as the most natural follow-on experiments. TC-ViTok scores low on TextVQA (25.8%), and the matched LoRA-576 control scores substantially higher (38.4%). This is expected behavior rather than a hidden strength of the method: OCR requires dense local coverage across the entire image, while our $4\times$ selected-token compression with k=144 deliberately discards most tokens and the positional prior documented in L9 concentrates selection on a subset of positions regardless of OCR needs. We do not present TC-ViTok as competitive on OCR-centric benchmarks. The LoRA-576 control itself remains below the 45–55% range of publicly released LLaVA-1.5 variants because our LoRA-SFT mix deliberately excludes OCR-VQA, TextVQA, DocVQA, and other OCR-heavy data (see Anticipated Questions Q1). The additional -12.6 pp gap from LoRA-576 to TC-ViTok on TextVQA isolates the OCR-specific cost of compressing to 144 selected tokens. On the OCR-free benchmarks (VQAv2, GQA, POPE) the LoRA-576 control reaches $76.38/70.55/86.46$, close to the publicly released LLaVA-1.5 numbers, confirming that the severe TextVQA degradation is specific to dense-text coverage rather than a general failure of the LoRA control.

(L7) E3 (zero-CLIP-query) marginally outperforms query-conditioned. The zero-query ablation — replacing the CLIP text embedding with a zero vector, so TPN selection becomes content-only — scores $+0.40$ VQAv2, $+0.33$ GQA, $+0.28$ POPE over the query-conditioned baseline (Table 11). We repeated the same E3 protocol on Adaptive v2 over 5K samples: baseline scores are 69.10 / 66.36 / 75.74 on VQAv2 / GQA / POPE, while zero-query scores are 69.66 / 66.20 / 76.15, giving deltas of $+0.56$ / -0.16 / $+0.41$ pp. Thus the no-degradation pattern persists on an independently continued checkpoint, although the GQA delta is slightly negative and well below the benchmark-scale effects in Table 6. This does not refute Hypothesis H2; it refines it. A subsequent from-scratch image-only run (Table 9) converged more slowly than the query-conditioned variant, indicating the CLIP query accelerates early training by enabling the diversity loss that stabilizes against entropy collapse — but is approximately redundant at convergence once the TPN's spatial-attention bias has formed (H2′). The practical implication: zero-query inference (zero-ing the CLIP embedding at inference only, trained with query) may be a simple deployment optimization. H1 and H3 remain hypothesized; the qualitative heatmaps in Fig. 10 are consistent with both but do not constitute proof. The selection-frequency analysis (Fig. 11) makes the same point geometrically: per-position selection probability has a strong position-specific heterogeneity with mild upper-region preference, while mean score is nearly uniform except for a localized upper-center dead zone, so the model is in practice a *content-conditioned ranker with stable per-position biases, including measurable low-utility allocation* (L9, M1–M3). We do not claim this is desirable. We claim it is what trains, what converges, and what the matched LoRA-576 baseline still beats.

(L8) The architectural ceiling. We were unable to improve

the warm-started 0.3177 Val-VQA in three independent fine-tuning attempts (Section 6.2); every continuation either regressed or plateaued. We interpret this as evidence of an architectural ceiling at k=144 for this encoder/LLM pair, not as a tuning failure. Pushing meaningfully below 0.30 Val-VQA likely requires changes at one of three levels: a multi-step compression curriculum ($576 \to 288 \to 144$), a larger LLM backbone whose marginal value of additional visual context is higher, or a heterogeneous token vocabulary (selected + summary) that the parameter-free merge does not yet implement. We flag this so that downstream work does not exhaust compute reproducing our fine-tuning results before tackling the structural axis.

(L9) Structured per-position selection prior with a localized score dead zone. The aggregate selection diagnostic (Fig. 11) reveals two stable properties of the converged TPN: strong per-position heterogeneity in selection frequency (P_{select} ranging 0–0.7, with a long tail of near-zero positions and mild upper-region preference), and a localized dead zone of ~ 16 patches in the upper-center where mean score is systematically depressed. Masking the dead-zone positions (rows 0–4, cols 6–11) at inference by setting their TPN scores to $-\infty$ before top-k selection improves exact-match VQA-score from $0.678 \to 0.700$ (Δ=+2.2 pp; 95% CI $[+0.4, +4.4]$ pp) over 500 GQA val samples on Final. Re-scoring the same saved predictions with the soft VQAv2-style answer-matching metric gives the same direction: $0.2260 \to 0.2333$ (Δ=+0.73 pp; 95% CI $[+0.13, +1.47]$ pp; with GQA's single canonical answer per question, the soft metric uses $\min(\text{matches}/3, 1.0)$ and is therefore on a lower absolute scale than exact match). Across 10 size-matched random rectangles that do not overlap the dead zone, random masking also often improves accuracy (mean Δ=+1.58 pp; range +0.6–+2.4 pp), and the dead-zone effect ranks at the 90th percentile (exceeding 9 of 10 size-matched random rectangles). Under the null hypothesis that dead-zone and random rectangles are exchangeable, the probability of this rank is $p = 0.0107$ (one-tailed binomial sign test, n=10, k=9, $H_0 : p$=0.5). The mean paired differential $\Delta_{\text{DZ}} - \Delta_{\text{rect}}$ over the 10 controls is only +0.62 pp, and most pairwise bootstrap intervals include zero. Despite the per-rectangle differential CIs being wide, the dead zone is the only region selected a priori (before M1 was run) by visual inspection of an aggregated diagnostic computed before the masking experiments, providing weak prior-registration evidence that we did not data-mine the result. We therefore cannot claim the gain is uniquely attributable to the dead zone; the safer conclusion is that the converged selector allocates a small but measurable fraction of its budget to low-utility positions, with the visually identified dead zone among the strongest such regions. The dead-zone bounds were chosen by visual inspection of Fig. 11(right) before the masking experiment was run, not by post-hoc optimization on M1. We also ran a held-out cross-dataset check: identifying low-score positions on 500 VQAv2 val samples recovers only a small subset of the GQA-derived region (rows 0–1, col 8; Jaccard 0.067 against rows 0–4, cols 6–11). Masking those VQAv2-derived positions on the same 500 GQA samples changes exact-match VQA-score from $0.678 \to 0.692$ (Δ=+1.4 pp; 95% CI $[-0.2, +3.2]$ pp). This does not validate the full GQA dead-zone box as a dataset-independent object; it is weakly consistent with a narrower upper-center low-utility subset. The low Jaccard is expected on first principles: the selection-frequency diagnostic measures which patches the selector ranks highly on a given image distribution, and VQAv2 and GQA have systematically different image and question distributions. The narrow upper-center subset that survives across both datasets (rows 0–1, col 8) is the dataset-invariant core, while the wider GQA-derived box reflects GQA-conditional extension. This is a deliberate revision from the stronger "attention sink" language: M1–M3 support a low-utility allocation observation, while the mechanism remains unisolated. Attention-sink dynamics, positional-encoding artifact, content priors from training-data statistics, and LoRA fine-tuning artifacts are all consistent with the observation.

We replicate the same dead-zone mask test on two related checkpoints and one independently retrained seed: Adaptive v2 improves from $0.676 \to 0.700$ (Δ=+2.4 pp; 95% CI $[+0.4, +4.4]$ pp), and v3 improves from $0.590 \to 0.612$ (Δ=+2.2 pp; 95% CI $[-0.6, +4.8]$ pp), while seed 7 decreases from $0.692 \to 0.682$ (Δ=−1.0 pp; 95% CI $[-2.8, +0.8]$ pp). Thus the dead-zone mask is not checkpoint-invariant: it is positive on the original checkpoint family, but does not replicate on the independently retrained seed. The v3 checkpoint also has a substantially lower baseline (0.590 vs. 0.678/0.676 for Final/Adaptive v2), so it leaves more headroom for any inference-time intervention. We therefore keep L9 as an observation about the Final checkpoint's converged positional prior and as evidence that low-utility allocation can occur, not as a universal dead-zone mechanism. Full sink-vs-register analysis (register-token-equipped CLIP encoders, explicit anti-sink regularization during training, multi-checkpoint selection-frequency replication) is left for future work. The broader scope statement is: TC-ViTok is best described as a *content-conditioned ranker with stable per-position biases*, not as a query-conditioned selector at this scale.

7. Limitations

We explicitly enumerate the limitations of this work to guide future follow-up studies.

This section enumerates *paper-level limitations* of the contributions and methodology. The L1–L9 entries embedded

in Section 6 are detailed empirical observations and "lessons learned" from the training-dynamics study; the items below (L-i through L-v) are top-level limitations of what the paper establishes. The two are distinct in scope and we keep them separate deliberately.

(L-i) Limited seed coverage. Table 6 numbers are reported for the seed-42 Final checkpoint, and Table 7 adds one independently retrained seed for the headline LoRA-576 comparison. This is still not a full multi-seed study. Binomial-approximation standard errors on the listed sample sizes are $\sigma \approx 0.16$–0.50 pp depending on the benchmark; reported deltas of ≥ 1.5 pp are above this noise floor, while smaller margins (e.g., the $+0.40$ pp E3 delta on VQAv2) are at or below it. The seed-7 M1 replication also changes sign, so L9 is explicitly not presented as a checkpoint-invariant mechanism.

(L-ii) Single-architecture taxonomy. The three failure modes of Sections 3.2–6.2 are observed in one (encoder, LLM, selector) triple: CLIP ViT-L/14, LLaVA-1.5-7B, and the TPN of Section 3. We do not establish that these pathologies occur in other encoder/LLM/selector combinations; we present them as testable hypotheses.

(L-iii) Compression cost remains. TC-ViTok is dominated by a matched LoRA-576 reference (no compression) by 4.1 / 3.4 / 7.4 points on VQAv2 / GQA / POPE. We do not close this gap; the paper's empirical content is the characterization of why, not a SOTA claim.

(L-iv) Dead-zone mechanism not isolated. M1–M3 show that masking a localized region of the selector's selection space improves downstream accuracy and ranks above most size-matched random rectangles, but the paired differentials are weak and the held-out VQAv2 diagnostic recovers only a small subset of the GQA-derived region. We do not establish what causes the region to be low-utility: attention-sink dynamics, positional-encoding artifact, content prior from training-data statistics, and LoRA fine-tuning artifact are all consistent with the observation. We present the observation; mechanism is future work.

(L-v) Two-point oscillation slope. The oscillation-period characterization (§6.4) rests on two finite-period measurements and one frozen-pooler control. We report this as a trend, not a scaling law; confidence interval on the slope is underdetermined with $n{=}2$ points.

8. Conclusion

We presented TC-ViTok, but the paper's central object is not a new benchmark number. It is a taxonomy of three coupled training pathologies that appeared in our pre-LLM trainable visual-token compression system when its selector was co-trained with LLM adapters: co-adaptation oscillation, BudgetPredictor collapse, and warm-start dependence. For each we provide a mechanistic explanation (Sections 3.2, 6.3, 6.2), an empirical signature, and an architectural or curricular fix — parameter-free token merge, a calibrated λ_b window, and a three-stage curriculum staged on a converged base.

On LLaVA-1.5-7B at $4\times$ selected-token compression and matched LoRA SFT, the resulting system scores **4.1 VQAv2 points** and **3.4 GQA points** below the same-LoRA full-token reference, and **7.4 POPE points** below it — a larger cost we interpret as evidence the system correctly drops question-irrelevant tokens even when those tokens contain detectable objects. We were unable to push Val-VQA meaningfully below the warm-started ceiling (Sec. 6.2, L8); we report this honestly because it sharply localizes where future architectural work is needed.

The most durable artifact, in our reading, is the failure-mode taxonomy and its predictions: that oscillation period increases as selector learning rate decreases in the measured trainable-pooler family (Sec. 6.4), that budget heads collapse outside the $\lambda_b \in [0.08, 0.15]$ window (Sec. 6.3), and that no fix for the from-scratch plateau exists at this LLaVA scale that we are aware of (Sec. 6.2). These are concrete, falsifiable claims about how trainable pre-LLM selectors behave; we expect downstream work to find both confirming instances and counterexamples, and we have tried to write the paper such that either outcome is informative.

M1–M3 test the convergent-prior finding directly: masking the upper-center dead zone improves VQA-score by $+2.2$ pp on Final and ranks at the 90th percentile of 10 size-matched random controls, but the sign is mixed across replications (positive on Adaptive v2 and v3, negative on seed 7). The result supports low-utility budget allocation as a failure mode that can appear in the trained selector, but it does not establish a checkpoint-invariant dead-zone mechanism or rule out generic redistribution away from other low-utility regions. The remaining claims are stated as falsifiable predictions: the oscillation trend is falsified by an otherwise identical trainable-pooler architecture where T_{osc} does not increase as η_{selector} decreases; the budget-collapse law is falsified by a stable k_{eff} outside $[0.08, 0.15]$ at the same LLM scale; and warm-start dependence is falsified by any from-scratch run that converges below 0.34 Val-VQA at LLaVA-1.5-7B scale without warm initialization.

Future work. (i) Multi-step compression ($576 \to 288 \to 144$), which we conjecture is the architectural axis along which the warm-started ceiling can move; (ii) RefCOCO, DocVQA, and ChartQA, where query-conditioning's value is hypothesized to be largest; (iii) the unmeasured λ_b Pareto points outside the collapse boundary, to give a full operating curve rather than the three-point one we report; (iv) generalization of the failure-mode taxonomy to Qwen-

VL, InternVL, and larger LLM backbones, which would test whether the training-dynamics laws we report are LLaVA-specific or generic to pre-LLM compression. We specifically predict that replacing the CLIP vision encoder with a register-token-equipped variant in the style of Darcet et al. (Darcet et al., 2024) would attenuate the localized low-score region in Fig. 11; this is a short, falsifiable test of the attention-sink mechanism that we do not run here.

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A. Anticipated Reviewer Questions

We pre-answer the questions we expect reviewers to raise.

Q1: Why does TC-ViTok underperform training-free methods like FastV and PruMerge? FastV (77.1% VQAv2 at k=144) and PruMerge (76.8%) are post-hoc compression methods applied to the publicly released LLaVA-1.5-7B, which itself was supervised-fine-tuned on a richer

mixture than ours (LLaVA-Instruct, VQAv2, GQA, OCR-VQA, TextVQA, A-OKVQA, RefCOCO, and others). Our matched LoRA-576 control — the same LoRA capacity and the same SFT mix as TC-ViTok Final, but without compression — scores 76.38% VQAv2, within 0.7 pp of FastV's 77.1%. The apparent FastV advantage over TC-ViTok (+4.8 pp) is therefore primarily attributable to the difference between our LoRA-SFT mix and the public LLaVA-1.5 SFT mix, not to the compression mechanism. Within the LoRA-SFT regime in which TC-ViTok operates, the actual compression cost is the $4.1/3.4/7.4$ pp gap to LoRA-576 that we report in the abstract.

We chose the restricted SFT mix deliberately, for two reasons: (i) the full LLaVA-1.5 SFT mix would require roughly $5\times$ our available training compute, and (ii) we wanted a clean control where TC-ViTok and LoRA-576 differ *only* in the presence of compression, so that the compression cost is not confounded with SFT-mix differences. We do not claim TC-ViTok beats training-free methods at absolute accuracy; we claim it is, to our knowledge, the first demonstration that a trainable query-conditioned pre-LLM selector can *converge* at LLaVA-1.5-7B scale, and we document the failure modes encountered during that demonstration (§6).

We also do not directly evaluate against QG-VTC (Li et al., 2025), the closest query-aware compressor: QG-VTC compresses *inside* the encoder via in-tower attention with the question, whereas TC-ViTok compresses *after* the encoder via a modular pre-LLM module operating on frozen encoder features. These are different points in the architectural design space; we position TC-ViTok in §2 accordingly.

Q2: How do you address the concern that the dead-zone region was identified and tested on the same 500 GQA samples? We ran the held-out check and report the result directly. Identifying low-score positions on 500 VQAv2 val samples with the same threshold recovers rows 0–1, col 8, which overlaps the GQA-derived rows 0–4, cols 6–11 box with Jaccard 0.067. Masking the VQAv2-derived positions on the original 500 GQA samples improves exact-match VQA-score from 0.678 to 0.692 (Δ=+1.4 pp; 95% CI $[-0.2, +3.2]$ pp). This is not strong cross-dataset validation of the full GQA box; it is weak evidence that a narrower upper-center subset is consistently low-utility. We therefore frame L9 as an observational low-utility allocation result, not as proof of a dataset-independent attention sink. The low overlap is expected if the selector's positional prior is partly conditioned by the image and question distribution; the cross-dataset survivor (rows 0–1, col 8) is the narrow dataset-invariant core, while the wider GQA box is a GQA-conditional extension. The main M1 effect is not an artifact of exact-match scoring: re-scoring the saved Final predictions with the soft VQAv2-style metric gives $0.2260 \rightarrow 0.2333$ (Δ=+0.73 pp; 95% CI $[+0.13, +1.47]$ pp).

Q3: Why only N=10 random rectangles for the M3 control? Each masked-eval run requires a full greedy generation pass through LLaVA-1.5-7B on 500 samples. N=10 was chosen to balance percentile resolution against runtime. We therefore report the 9/10 sign-test p=0.0107 (§6.7, L9) rather than relying on the percentile point estimate alone.

Q4: Is the 4.1/3.4/7.4 LoRA-576 gap a failure of the method? Yes. The method is dominated by a no-compression baseline. We do not hide this; we present the gap as the empirical quantity that motivates the failure-mode taxonomy in §6. The contribution is not that TC-ViTok beats the matched no-compression reference, but that it exposes where this family of pre-LLM trainable compressors loses accuracy.

Q5: Why is POPE the largest gap (-7.4 pp)? POPE evaluates per-object hallucination by asking whether a specific object is present in the image. Query-conditioned pre-LLM compression drops patches that the current question does not target, including patches containing detectable objects unrelated to the question. The POPE gap is therefore an expected cost of query-conditioned compression. The matched LoRA-576 control sees all 576 patches and does not face this constraint.

Q6: Why is the E3 zero-query finding considered a contribution rather than a flaw of the method? The TPN is architecturally a query-conditioned selector: the CLIP text embedding is a non-trivial input. The observation that the converged selector achieves equal or marginally better performance with that embedding zeroed is an unexpected property of joint optimization at this scale, and directly tests whether the architecture's design intent is realized by training. Our result is that it is not, at least at LLaVA-1.5-7B with k=144.

B. Reproducibility

Code. Source code, training scripts, and evaluation scripts will be released at an anonymous URL during review or publicly upon acceptance, consistent with the double-blind policy. The held-out validation, M1/M3 controls, and VQAv2-score post-processing are all implemented as standalone scripts under `scripts/`.

Checkpoints and run metadata. Each checkpoint stores its model state and training configuration. The main reported rows use the Phase 1 Final checkpoint at 80K steps, the Adaptive v2 continuation, the Phase 1 v3 checkpoint, and the LoRA-576 control described in Table 6. We do not claim multi-seed benchmark averages; all reported benchmark rows are single-seed unless explicitly stated otherwise.

Hardware and runtime. Training used RTX 6000 Ada

GPUs as summarized in Table 5. Masked 500-sample ablations (M1/M3) run on one high-memory GPU and require one baseline pass plus one generation pass per mask condition. The script outputs per-sample paired scores so bootstrap intervals can be recomputed exactly.

C. Ethics and Negative Impact

TC-ViTok is an efficiency-oriented VLM compression method and does not introduce new data collection or human-subject risks. The main negative impact is deployment risk from selectively dropping visual tokens: query-conditioned compression can remove objects irrelevant to the current question, which is undesirable for safety-critical detection, OCR, or auditing workflows. We therefore recommend using no-compression or task-specific validation for those settings.



Figure 10. **TPN selection on six GQA val_balanced examples.** Each row: raw image (left), top-k=144 selection overlay (right). The overlay tints only selected patches with warm colors; alpha is modulated by the per-image min-max normalized TPN score. Per-image normalization saturates the colormap on whichever positions rank highest *within that image*, which can produce the visual impression of a regional bias. The aggregate-over-500-samples diagnostic (Fig. 11) shows the underlying pattern is heterogeneous and only mildly regional — much weaker than per-sample visualization suggests. We include these examples to make per-sample selection legible, not as evidence of question-conditioned focus, which the aggregate analysis refutes.

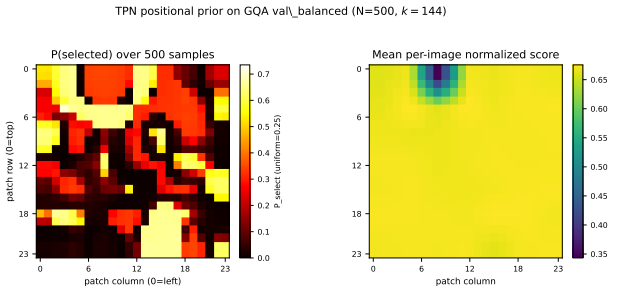


Figure 11. **Aggregate TPN selection diagnostics on 500 GQA val_balanced samples. How to read:** compare the left panel to the uniform baseline $P_{\text{select}}{=}0.25$, then check whether the right panel has localized score depressions that explain the frequency map. *Left:* per-position selection frequency P_{select}. Under fully content-driven selection at $k{=}144$, the uniform baseline is $144/576{=}0.25$ (per-position). The measured distribution is strongly heterogeneous (range 0–0.7, with 14% of positions in a near-zero tail $P_{\text{select}} < 0.02$): specific positions are selected on $> 60\%$ of samples while others are selected on $< 5\%$. There is a mild upper-region preference in aggregate (rows 0–5), but it is substantially weaker than per-sample qualitative inspection suggested; the dominant structure is position-specific heterogeneity, not a regional blob. *Right:* per-image min-max normalized score, averaged across samples, is approximately uniform ($\bar{s} \approx 0.65$) *except* for a localized $\sim 4{\times}4$-patch "dead zone" in the upper-center ($\bar{s} \approx 0.35$, rows 0–4, cols 6–11) that the TPN systematically deprioritizes regardless of content. We conjecture this corresponds to an attention-sink position learned during joint training; testing this with register tokens (Darcet et al., 2024) or sink masking is a direct follow-on. Together with the E3 ablation (§6.7, L7), these aggregate statistics give a geometric account of the convergent TPN behavior that per-sample qualitative inspection (Fig. 10) cannot provide.

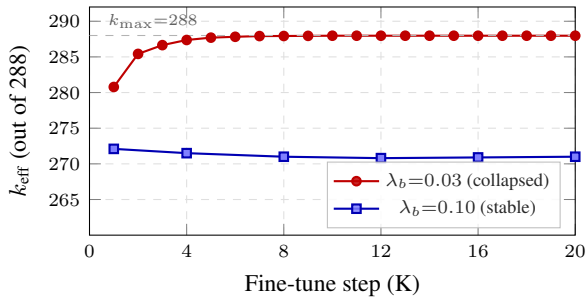


Figure 12. **BudgetPredictor collapse vs. stable regime (fine-tune from warm-started checkpoint).** At $\lambda_b{=}0.03$, the head saturates against $k_{\max}{=}288$ within 10K steps and stays pinned; val-VQA simultaneously regresses from 0.3726 to 0.3898 (text in Section 6.3). At $\lambda_b{=}0.10$, the head holds $k_{\text{eff}} \approx 271$ for the full 20K-step horizon. The collapse boundary lies between these two values; the recommended operating window is $\lambda_b \in [0.08, 0.15]$.

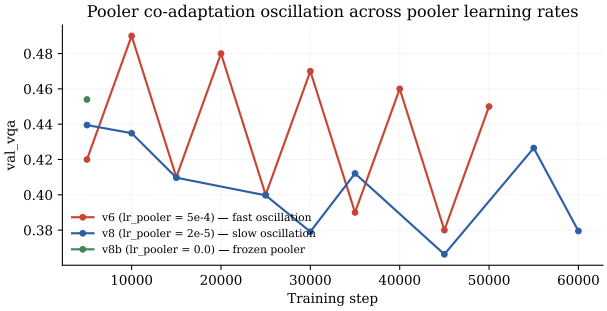


Figure 13. **Co-adaptation oscillation across pooler learning rates.** Validation VQA accuracy on the GQA val_balanced subset for three Phase 1.5 trainable-pooler variants. **Top**: $\eta_{\text{pooler}}{=}5{\times}10^{-4}$ (v6), period $\approx$5K steps. **Middle**: $\eta_{\text{pooler}}{=}2{\times}10^{-5}$ (v8), period $\approx$15K steps; the trajectory is well-behaved through step 30K and deteriorates from step 35K onward. **Bottom**: $\eta_{\text{pooler}}{=}0$ (v8b, frozen pooler), no oscillation but the pooler outputs become fixed noise the LLM cannot use. The period scales as $1/\eta_{\text{pooler}}$, the empirical signature of the GAN-style co-adaptation loop diagnosed in Section 3.2.